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Deep learning-based multi-criteria recommender system for technology-enhanced learning

Latifat Salau[1]✉, Hamada Mohamed[2], Yunusa Simpa Abdulsalam[3] & Hassan Mohammed[4]

Multi-Criteria Recommender Systems (MCRSs) improve personalization by incorporating multiple user preferences. However, their application in Technology-Enhanced Learning (TEL) remains limited due to challenges such as data sparsity, over-specialization, and cold-start problems. Traditional techniques, such as Singular Value Decomposition (SVD) and SVD + +, struggle to effectively model the complex interactions within multi-criteria rating data, leading to suboptimal recommendations. This paper introduces a hybrid DeepFM-SVD + + model, which integrates deep learning and factorization-based techniques to improve multi-criteria recommendations. The model captures both low-order feature interactions using factorization machines and high-order dependencies through deep neural networks, enabling more adaptive recommendations. To evaluate its performance, the model is tested on two multi-criteria datasets: ITM-Rec (TEL domain) and Yahoo Movies (non-TEL domain). The experimental results show that DeepFM-SVD + + consistently outperforms the traditional techniques across multiple evaluation metrics. The findings highlight significant improvements in accuracy, demonstrating the model's effectiveness in sparse datasets and its generalization across domains. By addressing the limitations of existing MCRS techniques, this study contributes to advancing personalized learning recommendations in TEL and expands the applicability of deep learning-based MCRS beyond educational contexts.

Keywords Multi-criteria recommender systems, Technology-enhanced learning, Recommender systems, Deep learning

Recommender systems (RSs) have evolved significantly since their early applications in the mid-1990s. This was driven by the need to filter vast amounts of digital content and tailor suggestions to individual user preferences. These systems analyze user behavior, contextual data, and item interactions to generate personalized recommendations. RSs serve as essential tools representing a worthwhile set of resources that make available suggestions most likely to interest users[1]. The rapid growth of online services such as e-commerce, e-learning, and e-government, along with others, in conjunction with the large volume of items in their databases, makes RSs necessary for recommending interesting items not yet seen by users[2].

Early RSs primarily analyzed explicit user preferences, but these systems struggled with scalability and personalization as digital content grew exponentially. As a result, collaborative filtering techniques introduce user-item interactions to infer recommendations. Methods such as matrix factorization played a crucial role in improving recommendation accuracy. By the 2000s, data sparsity and cold-start problems were being addressed using hybrid models that combined collaborative and content-based filtering. The continuous growth in the complexity of user behavior and large-scale datasets in the 2010s led to the adoption of deep learning techniques, allowing RSs to capture intricate patterns in implicit feedback, contextual cues, and dynamic user preferences. Modern RSs now leverage deep neural networks and hybrid architectures to refine recommendations, significantly enhancing accuracy, user engagement, and retention across various applications.

Multi-criteria recommender systems (MCRS) represent an advanced evolution of traditional RSs that provide more personalized and nuanced recommendations to users. While traditional RSs rely on a single score or preference to make recommendations, MCRS considers the user's preference across multiple criteria, allowing for a more tailored experience. With MCRS, users can find items that meet their specific needs and priorities rather than just overall ratings or single-dimensional features. MCRS considers additional information, such as multiple criteria ratings on user preference, to improve the quality of recommendations by considering different

[1]Department of Computer Science and Engineering, African University of Science and Technology, Abuja 900109, Nigeria. [2]Software Engineering Lab, University of Aizu, AizuwakamatsuFukushima 965-8580, Japan. [3]College of Computing, University Mohammed VI Polytechnic, 43150 Ben Guerir, Morocco. [4]Department of Software Engineering, Bayero University, Kano 700241, Nigeria. ✉email: ladbdulsalam@aust.edu.ng

attributes of the items. The versatility of MCRSs has led to widespread adoption in real-world applications spanning multiple domains. In the entertainment industry, platforms like Yahoo! Movies utilize MCRSs to refine movie recommendations based on aspects such as *direction, action, story, and visuals*. Similarly, TripAdvisor employs MCRSs for hotel and travel recommendations, factoring in aspects like *location, cleanliness, and amenities*. Platforms like OpenTable incorporate multi-criteria ratings in the restaurant industry to enhance dining recommendations. Figure 1 provides an example from OpenTable, illustrating how MCRSs optimize user choices by considering multiple evaluation criteria. The OpenTable offers recommendations for a restaurant based on aspects like food, service, ambience, value, and noise.

Traditionally, RSs are based on clustering, nearest neighbor, and matrix factorization methods[3]. However, deep learning has yielded tremendous success in recent years across multiple domains, from image recognition to natural language processing[4]. RSs have also benefited from the success of deep learning. Deep learning is a subfield of machine learning that makes use of artificial neural networks with multiple layers to learn complex patterns or representations in data. These layers progressively extract higher-level features from the raw input, allowing the model to make more accurate predictions or decisions. Deep learning techniques have been proven to enhance the performance of RSs by improving the accuracy of recommendations because they can exploit complex and non-linear user-item relationships[5]. By integrating multi-criteria decision-making techniques with modern deep learning approaches, MCRSs continue to push the boundaries of personalized recommendation systems, making them a powerful tool for improving decision-making in education, entertainment, and e-commerce.

Educational RSs have emerged as a technology-enhanced learning method[6]. There has been an increase in the growth of academic resources, which makes it quite difficult for traditional search engines to meet the requirements related to learners' online searches for information about educational products and services during their learning process[7]. In technology-enhanced learning, RSs are used to find and suggest suitable learning objects to the learner, and a learning object (e.g., a problem) has several categories that indicate topics or fields[8]. Many educational RSs have been proposed and deployed in learning, teaching, and academic advising[9]. The relevance of recommending educational resources will help improve students' learning processes[10] by providing them with personalized, suitable, and helpful information.

Researchers and industries have used different methods to build RSs that intelligently recommend learning or teaching materials to users[3]. Collaborative, content-based, and hybrid filtering techniques are used to build RSs for technology-enhanced learning[6]. Collaborative filtering techniques have provided the most accurate RSs as they learn from all the existing implicit or explicit feedback about how users rate or use items[11]. However, over-specialization, data sparsity, and cold-start problems are the major drawbacks of these traditional techniques[12]. Federated recommender systems have emerged as a promising solution to these challenges, leveraging decentralized learning while preserving user privacy[13,14]. The most robust approach to capturing complex relations is to use deep learning techniques[15]. Federated learning combined with deep reinforcement learning has also been explored for improving communication efficiency in distributed recommendation models[16], as well as many other deep learning techniques. Deep learning solutions in collaborative filtering RSs are majorly classified as Deep Factorization Machine (DeepFM)[17] and Neural Collaborative Filtering (NCF)[18]. DeepFM architecture simultaneously learns low and high-order feature interactions from the raw features of the input. The NCF is based on dual neural networks that simultaneously process user and item information[11]. DeepFM has proven to perform better than NCF. DeepFM achieves an accuracy gain of 0.48% (Company dataset) and 0.33% (Criteo dataset) over NCF[19].

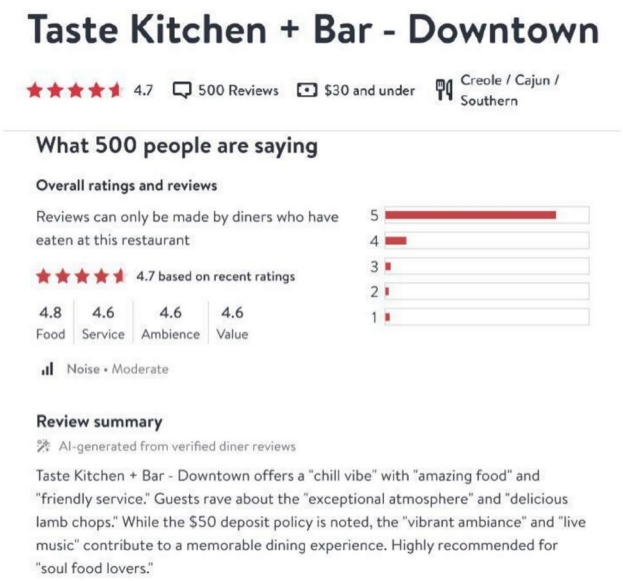

Fig. 1. MCRS example from OpenTable.com.

Despite the growing adoption of deep learning techniques in general RSs, their application in MCRSs for TEL remains underdeveloped. Existing TEL recommendation systems primarily rely on single-criteria approaches, which fail to account for learners' multidimensional preferences. Some works have attempted to integrate multi-criteria ratings, but these methods typically use matrix factorization techniques that struggle with cold-start issues.

While DeepFM and SVD/SVD + + have shown effectiveness in general recommender domains, their direct application to TEL remains unexplored. TEL presents distinct challenges, including heterogeneous learner preferences, complex learning environments, and the need for adaptive resource recommendations. Existing approaches in TEL fail to jointly model multi-criteria dependencies, often treating them as independent features rather than interrelated components influencing overall learning experiences.

To bridge this gap, this study proposes a DeepFM-SVD + + model optimized for TEL environments. Unlike traditional collaborative filtering approaches, our model leverages deep learning to capture both low-order and high-order interactions between multi-criteria ratings. This overcomes the limitations of prior MCRS techniques, improving recommendation accuracy, personalization, and generalization across TEL datasets.

Furthermore, a critical gap in MCRS research is the lack of optimized techniques for computing an aggregated overall rating from multiple criteria. Traditional aggregation methods rely on static weighting schemes or linear combinations, which do not fully capture non-linear relationships between multi-criteria ratings. To address this, we introduce an aggregation function that integrates deep learning-based factorization models (DeepFM) with Singular Value Decomposition (SVD + +), enabling the system to adjust rating weights based on learned patterns dynamically. Through this approach, we advance multi-criteria recommendations in TEL by providing a more adaptive and personalized framework that effectively models complex learner interactions, mitigates data sparsity, and enhances recommendation precision.

To summarize, this study makes the following key contributions to multi-criteria recommendations in TEL:

(1) Advancing multi-criteria recommendation in TEL: Most existing MCRS research focuses on e-commerce, tourism, or entertainment domains, with limited attention to personalized learning environments. This study addresses this gap by introducing a novel DeepFM-SVD + +/SVD model, which integrates deep learning with factorization machine techniques to optimize multi-criteria relationships in TEL, enabling more adaptive and personalized recommendations.
(2) Enhanced predictive accuracy through hybrid modeling: Unlike traditional matrix factorization-based methods (SVD and SVD + +), our model captures both linear and high-order non-linear interactions between multi-criteria ratings, leveraging deep neural networks for more expressive feature learning. A rigorous comparative analysis against SVD and SVD + + demonstrates significant improvements in accuracy, ranking performance, and robustness, particularly in sparse datasets. Our experiments validate the model's ability to mitigate cold-start issues and enhance recommendation precision in multi-criteria settings.
(3) Generalization across domains: Our proposed model is applied to two datasets:

    - The ITM-Rec multi-criteria dataset to evaluate its relevance to TEL.
    - A multi-criteria Yahoo Movies dataset to highlight the model's generalization to non-educational domains.

Paper organization

This paper is structured into six sections, including the introduction. Section "Related work" introduces related work on MCRS in deep learning and technology-enhanced learning. Section "Preliminary" briefly introduces our traditional RSs technique and DeepFM and its application in MCRS. Our proposed MCRS framework is presented in section "Experiment". Section "Results and discussion" presents the result and discussion. Finally, section "Conclusion and future work" presents the conclusion and scope for future work.

Related work

Deep learning-based RSs in the educational sector are still in their early stages, with limited exploration of MCRSs in TEL. While the literature on MCRSs has predominantly focused on domains such as movies and tourism, their application in TEL remains unexamined. Most studies in TEL rely on single-criterion RSs, which, although effective in enhancing learning outcomes, fail to account for the multi-dimensional nature of learner preferences, such as engagement, content relevance, and difficulty level. Given the nascent state of MCRS research in TEL, this study aims to bridge this gap by employing deep learning techniques to model and optimize multi-criteria relationships, addressing the unique challenges associated with TEL. To provide a comprehensive overview of related work, we have structured the literature review into two key areas.

The first area, Deep Learning in Multi-Criteria Recommender Systems, examines recent advancements in applying deep learning to MCRSs. By analyzing existing approaches, we identify how deep learning methods have been leveraged to enhance recommendation accuracy and personalization in multi-criteria settings.

The second area, Multi-Criteria Recommender Systems in Education, explores previous studies on MCRSs, specifically within educational contexts. This section highlights various multi-criteria approaches used for learning resource recommendations and assesses their limitations in effectively capturing the diverse and multi-dimensional needs of learners. By identifying these gaps, we establish the motivation for our proposed model, which seeks to improve personalized recommendations in TEL through deep learning-based multi-criteria optimization.

By structuring the literature review in this manner, we provide a broad yet focused analysis of the MCRS landscape while emphasizing specific gaps in TEL-based MCRSs that this study aims to address.

Deep learning-based multi-criteria recommender systems

Several researchers have used deep learning architectures to improve the accuracy and performance of MCRSs. Son-Lam and Quang-Hung[20] proposed a novel approach that relied on deep learning for context-aware MCRS. The authors used deep neural network models to learn the aggregation function and predict the context-aware multi-criteria ratings. They conducted experiments on a real-world travel dataset. The experimental results based on the effectiveness of the recommendation showed that their proposed approach significantly outperformed the state-of-the-art methods. Yong and David[21] proposed a novel multi-criterion ranking framework that can directly produce a ranking score for an item candidate. The author utilized independent neural matrix factorization and multi-output neural matrix factorization models as baseline approaches and tried incorporating the multi-criteria ranking based on Pareto ranking into these baselines. The experimental results demonstrated the effectiveness of the proposed approach by evaluating the top-N recommendations over multiple real-world datasets. Qusai[5] proposed a deep autoencoder-based multi-criteria (AEMC) recommendation algorithm to exploit non-trivial, non-linear, and hidden relationships between users and items. The author aims to use the proposed approach to learn and model the user preferences and improve the accuracy of the recommendation. The experimental result revealed that the author's proposed AEMC recommendation algorithm significantly performed better than the state-of-the-art baseline algorithm based on the recommendation accuracy. Hassan et al.[22] proposed an adaptive linear neuron (Adaline), which was trained using delta rule and asymmetric singular value decomposition algorithms to examine the performance of multi-criteria RSs. The authors used the aggregation function approach, where the Adaline model was integrated with the traditional single rating asymmetric SVD. The experimental results showed an improved accuracy using the author's proposed approach compared to previous works. Based on the experimental result, the authors also suggested a strong association between artificial neural networks and the modeling approaches of multi-criteria recommendation techniques. Nour et al.[23] proposed a fused deep neural network and matrix factorization for multi-criteria collaborative filtering RSs. Their model consisted of a fused matrix factorization model and a deep neural network to predict the overall rating. The authors conducted several experiments, showing that their proposed approach significantly outperformed other methods. Based on the results, the authors established that multi-criteria collaborative filtering RSs can be enhanced using different deep learning techniques or more complex models.

All the above studies highlighted the importance of deep learning in MCRSs and how deep learning has been proven to achieve better performance. However, there are few studies on applying deep learning techniques in multi-criteria RSs for technology-enhanced learning. Our proposed approach assesses the significance of a deep learning technique in improving the performance of MCRSs in technology-enhanced learning. We suggested using a hybrid deep learning model that combines factorization machines and deep neural networks. Also, DeepFM is more advantageous than other deep learning models because of its flexibility and versatility, as it can be used in various settings, and it is easy to train and deploy compared to other deep learning models.

Multi-criteria recommender system for technology-enhanced learning

Significant improvements have been made in MCRSs, such as those applied in domains like entertainment, tourism, and retail. However, their focus primarily lies in addressing domain-specific challenges unrelated to TEL. For instance, research works in tourism prioritize context-aware recommendations based on location and trust, whereas MCRSs in movies emphasize entertainment-related criteria like acting or visuals. These approaches, while effective in their respective areas, fail to address the unique challenges of TEL, including heterogeneous learner preferences, data sparsity, and cold-start problems. Single-criterion RSs have been widely used in TEL to improve learning outcomes but are inherently limited in capturing the multi-dimensional nature of learner preferences. Existing MCRSs, while promising, primarily rely on traditional collaborative filtering techniques, which struggle with scalability and fail to model complex, non-linear relationships between user and item features.

MCRSs in technology-enhanced learning are continuously growing. Few works have been carried out in this area. Manouselis et al[24] investigated three educational real-world multi-criteria datasets using a collaborative filtering algorithm. The authors focused on a multi-criteria collaborative filtering algorithm supporting the recommendation service within a web portal for organic and sustainable education. Experimental results from this study indicated the importance of the frequent experimental investigation of an RS's various design options and the need to explore adaptive implementations in real-life RSs. Tomohiro and Yutaka[8] proposed a learning path RS that employs a sequential model and recurrent neural network to a learner's ability chart. The authors proposed an ability chart that displays the learner's and the user's objective scores for building a personalized RS. The authors evaluated the proposed approach and the recommended problems for each sample user, and they concluded that their proposed system could be functional for other e-learning systems. Yong et al.[25] proposed the integration of context-aware and multi-criteria decision-making in educational learning. The authors performed experiments using an educational dataset, and the results revealed that their proposed approach could help produce more accurate recommendations. They also extended their experiments on a tourism dataset to demonstrate that their proposed approach can be generalized to other domains. Pengcheng and Li [26] proposed a personalized RS based on users' preference behavior data analysis to analyze the online education recommendation model. The authors used a k-means algorithm for user clustering based on the user's criteria preference and grouped users with similar preferences in the same category. The experimental results of their proposed approach showed that the multi-criteria recommendation technique using user distance similarity worked best, and they concluded that a personalized recommendation algorithm based on user preference could effectively improve recommendation quality.

So far, there are very few contributions to MCRS in the educational domain. However, since most educational RSs are implemented using traditional techniques, there is a limit to the information that can be accessed to provide an accurate recommendation. Since conventional techniques also have limitations, such as scalability,

sparsity, and cold-start problems, it is essential to develop a system that tries to solve these problems. Unlike existing methods that rely solely on collaborative filtering or focus on unrelated domains, the proposed aggregation function-based DeepFM-SVD++model addresses the unique requirements of TEL by combining deep learning and collaborative filtering to optimize multi-criteria relationships.

Preliminary

This section provides an overview of the traditional RSs techniques (SVD and SVD++), the multi-criteria recommendation technique, and the DeepFM framework.

Collaborative filtering overview

The collaborative filtering technique is one of the most popular and powerful recommendation techniques that suggest items to users based on the ratings of other users. It performs well in complex object recommendations (such as movies, music, and products), which are seen to be autonomous from any machine-readable prediction of the items recommended[27]. There are two main types of collaborative filtering: user-based and item-based. The user-based collaborative filtering recommends items to users based on the ratings of other users with similar preferences. The item-based collaborative filtering recommends items to users based on the ratings of other items that the user has rated. Different algorithms are used for collaborative filtering techniques, such as factorization machines, nearest neighbors, SVD, SVD++, etc.

Traditional recommender systems techniques (SVD and SVD++)

SVD is a method derived from linear algebra that is frequently used as a dimensionality reduction technique in machine learning. SVD has scored high in accuracy and perceived personalization[28]. SVD in RSs is a matrix factorization model where vectors represent users and items in a latent factor space[2]. In SVD, every user u is associated with an n-dimensional latent vector $p_u \in \mathbb{R}^n$, and every item i is associated with a vector $q_i \in \mathbb{R}^n$, thus, to predict $r_{ui}\prime$ between u and i is given as:

$$r_{ui}\prime = b_{ui} + q_i^T p_u \tag{1}$$

$$b_{ui} = (\mu_u - \mu) + (\mu_i - \mu) + \mu \tag{2}$$

where, b_{ui} is the baseline predictor for normalizing $r_{ui}\prime$ and the value of b_{ui} between u and i is calculated using the overall average rating μ. The regularized squared error is minimized to estimate all the unknowns.

$$\sum_{r_{ui} \in R_{train}} (r_{ui} - r_{ui}\prime)^2 + \lambda \left(b_i^2 + b_u^2 + \|q_i\|^2 + \|p_u\|^2\right) \tag{3}$$

Stochastic gradient descent was used to perform the minimization[29] as shown in Eqs. 4a, b, c, and d.

$$b_u \leftarrow b_u + \gamma \left(e_{ui} - \lambda b_u\right) \tag{4a}$$

$$b_i \leftarrow b_i + \gamma(e_{ui} - \lambda b_i) \tag{4b}$$

$$p_u \leftarrow p_u + \gamma(e_{ui} \cdot q_i - \lambda p_u) \tag{4c}$$

$$q_i \leftarrow q_i + \gamma(e_{ui} \cdot p_u - \lambda q_i) \tag{4d}$$

where $e_{ui} = r_{ui} - r_{ui}\prime$.

SVD++is a family of SVD that has proven to be more efficient than traditional SVD by achieving better accuracy. SVD++takes into account implicit feedback information based on SVD, i.e., it adds $y_j \in \mathbb{R}^f$ for each item, and these item factors are used to describe the characteristics of the item, regardless of whether it has been evaluated[30]. In SVD++, the user's factor matrix is modeled such that a better user bias can obtain more information to predict the value of $r_{ui}\prime$. The prediction of an item I to user u in SVD++is given as:

$$r_{ui}\prime = \mu + b_u + b_i + q_i^T \left(p_u + |I(u)|^{-\frac{1}{2}} \sum_{j \in I(u)} y_j\right) \tag{5}$$

where μ is the global average rating, b_u and b_u are user and item biases, q_i and p_u are the latent factor vectors for item i and user u, respectively, and $|I(u)|$ represents the total number of items interacted with by user u. The additional term y_j captures implicit feedback, modeling user preferences based on interactions such as views, clicks, or purchases. By incorporating both explicit and implicit feedback[31], SVD++provides a more comprehensive user representation, leading to higher accuracy and better personalization. This makes it useful in scenarios with sparse rating data, addressing the cold-start problem and improving recommendation quality compared to the traditional SVD.

Deep factorization machine

Deep learning-based RSs have succeeded in various domain applications, such as computer vision, image processing, natural language processing, RSs, etc. The strength of deep learning models in discovering hidden features has enabled the production of practical solutions to the limitations of conventional RSs, such as accuracy,

scalability, and sparsity[5,18]. DeepFM is a deep learning technique that combines Deep Neural Networks (DNN) with Factorization Machines (FM) to solve recommendation and prediction problems. DeepFM was proposed in 2017 in a paper[17] by Huawei researchers. DeepFM is a hybrid model that seeks to capture low-level feature interactions through factorization and high-level feature interactions through DNN. DeepFM combines the power of FM for recommendation and deep learning for feature learning in a neural network architecture[17].

DeepFM architecture

DeepFM integrates FM and DNN components within a parallel architecture to effectively model both low-order and high-order feature interactions. First, the input layer receives user u and item i features, which are then transformed into latent feature vectors by an embedding layer. These latent vectors are subsequently fed in parallel into the FM and DNN components for further processing. The FM component calculates the pairwise interactions between all the latent features, which is done by multiplying the latent features together and taking the dot product of the results. The output of the FM component is a single value representing the contribution of the low-order feature interactions to the final prediction. Let $x \in \mathbb{R}^n$ represents the feature vectors of one sample, and y represents the corresponding label, which can be real-valued[19]. The FM component is calculated as follows:

$$y\prime(x) = \omega_0 + \sum_{i=1}^{n} \omega_i x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} \langle v_i, v_j \rangle x_i x_j \tag{6}$$

where $\omega_0 \in \mathbb{R}$ is the global bias; $w \in \mathbb{R}^n$ represents the weights of the ith variable; $V \in \mathbb{R}^{n \times k}$ denotes the feature embeddings; k is the dimensionality of latent factors, and $\langle v_i, v_j \rangle$ model the pairwise interaction between the ith and *jth* feature. Figure 2 below shows the FM architecture.

The Deep component is a Multi-Layer Perception (MLP). MLP takes the latent features as input and outputs a single value by capturing the data's high-order feature interactions and nonlinearities. Let $e_i \in \mathbb{R}^k$ represents the latent feature vector of the ith field, which is the embedding of the ith field as shown below:

$$z^{(0)} = [e_1, e_2, \ldots, e_m] \tag{7}$$

where m is the number of fields. $z^{(0)}$ is then fed into the DNN as shown below:

$$z^{(l+1)} = \sigma\left(W^{(l)} z^{(l)} + b^{(l)}\right) \tag{8}$$

where σ is the activation function, and l is the depth of the layer. $z^{(l)}$, $W^{(l)}$ and $b^{(l)}$ denotes the output, weight matrix, and bias of the *lth layer*, respectively. The DNN architecture is shown in Fig. 3 below.

The output of the FM and DNN components is summed up to produce the final prediction [19]. Let $y\prime_{FM}$ represent the output of the FM component, $y\prime_{DNN}$ denote the output of the DNN component and σ represent the sigmoid function. The final prediction of DeepFM is calculated as follows:

$$y\prime = \sigma(y\prime_{FM} + y\prime_{DNN}) \tag{9}$$

It is important to note that the FM and DNN components share the same feature embedding, as shown in Fig. 4 above. This makes the FM component unique, as it learns low-order and high-order feature interactions from raw features.

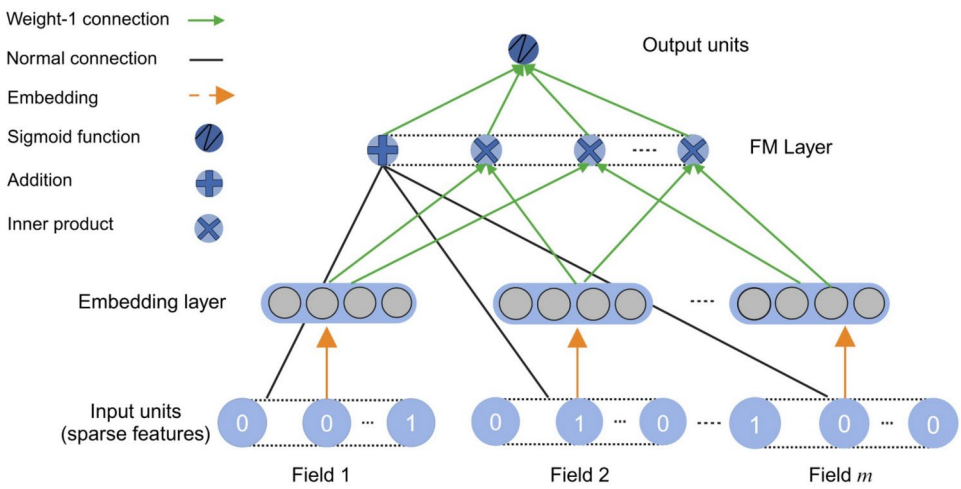

Fig. 2. The FM architecture.

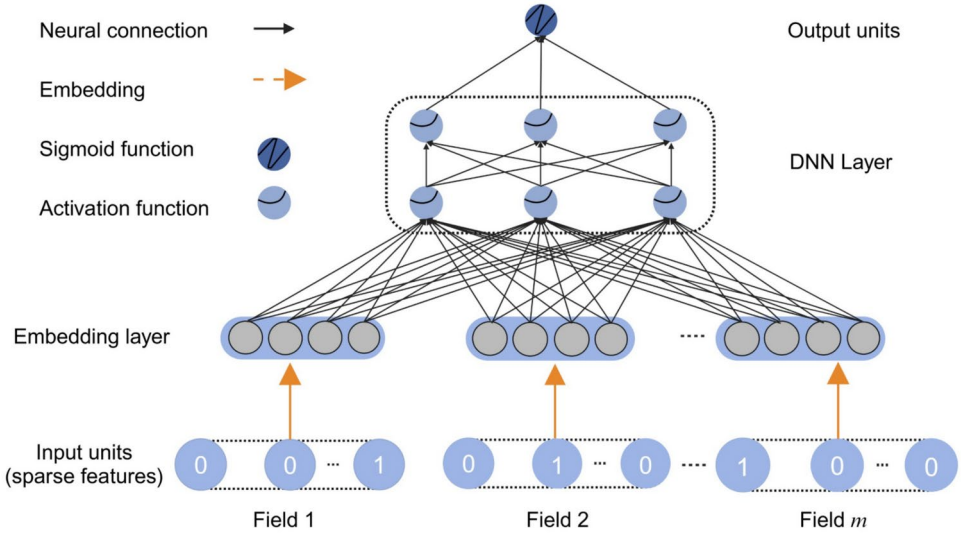

Fig. 3. The DNN architecture.

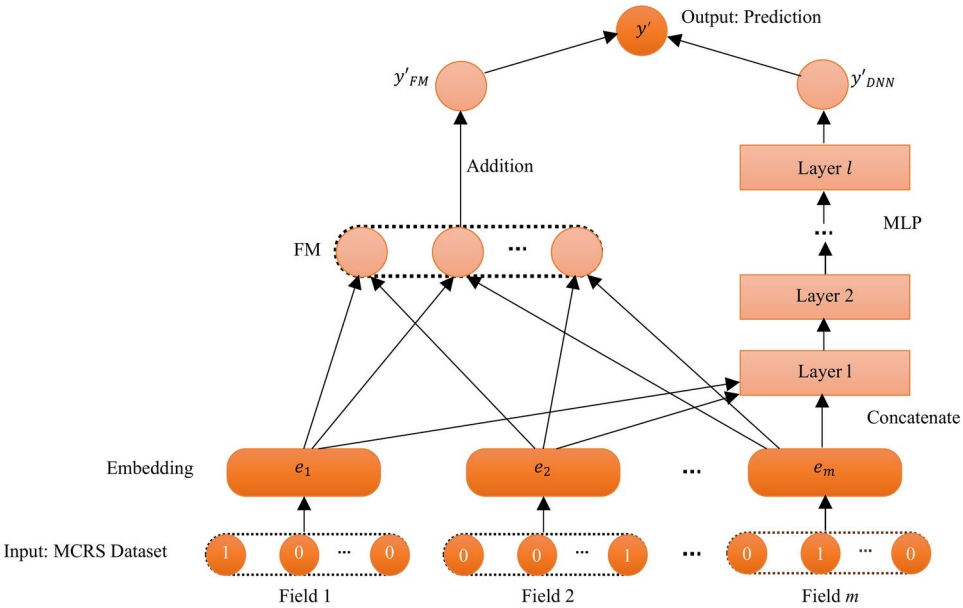

Fig. 4. The deepFM architecture.

Multi-criteria recommender systems

MCRSs are a powerful tool that can improve the accuracy of predictions and the quality of recommendations. Unlike the traditional RSs, which consider single criteria rating r, usually referred to as the overall rating, to represent the degree to which users u are interested in item I, MCRSs consider multiple criteria $r_0, r_1, r_2, \ldots, r_n$, for each criterion k ($k = 1, \ldots, n$), where r_0 is equivalent to the traditional RSs r. Thus, the MCRSs utility function f of recommending an appropriate item $i \in Items$ to user $u \in users$ is given as follows:

$$f : u \times i \rightarrow r_0 \times r_1 \times r_2 \times \cdots \times r_n \tag{10}$$

MCRSs can make more personalized recommendations that are more likely to be appreciated by users. It can also be implemented using traditional recommendation techniques such as collaborative filtering, content-based filtering, hybrid-based techniques, etc.[32]. Generally, RSs techniques can be categorized into two based on the formation of the utility function[33]: The heuristic-based and the model-based approach. The heuristic-based approach makes predictions based on the similarities between the user's observed data. There are primarily two fundamental principles of a heuristic-based approach: the user-based and item-based techniques. The model-based technique uses machine learning algorithms or statistical methods to build predictive models to learn

Dataset	Domain	Multi-criteria aspect	Size	Purpose
ITM-Rec	TEL	App, Data, Ease, Overall rating	5,230 ratings	Primary dataset for the proposed MCRS for TEL
Yahoo Movies	Entertainment	Direction, Action, Story, Visuals, Overall rating	62,156 ratings	Generalization to non-TEL domains

Table 1. Dataset overview.

UID	UserID	ItemID	App (c_1)	Data (c_2)	Ease (c_3)	Overall (c_0)	Class	Semester	Lockdown
1147–31-DA	1147	31	4	4	4	4	DA	Fall	PRE
249–50-DA	249	50	4	5	4	5	DA	Fall	PRE
4976–66-DM	4976	66	2	1	2	3	DM	Fall	POS
985–65-DA	985	65	5	4	3	4	DA	Fall	PRE
1275–26-DA	1275	26	1	2	2	1	DA	Spring	PRE

Table 2. Sample dataset for ITM-Rec rating.

user behavior and make suitable predictions. These predictive models include multilinear SVD, probabilistic modeling, and aggregation functions. The aggregation function model is the most used model among all the models. The aggregation function model assumes that the overall rating is an aggregate of the multi-criteria ratings[12,28].

Experiment

This section presents a conceptual summary of the proposed approach. It discusses the dataset used, the experimental framework of the approach, and how our system was modeled.

Datasets

To evaluate the performance and how well the proposed DeepFM-SVD + + model can be generalized, we used two datasets that cater to different recommendation system domains:

(1) ITM-REC dataset: This dataset focuses on multi-criteria ratings relevant to the field of TEL. It includes ratings on various aspects such as App, data, and ease. These criteria align closely with the challenges in personalized learning environments, making it ideal for validating the model's application in TEL contexts. The dataset is particularly useful for understanding the optimization of multi-criteria relationships.
(2) Yahoo movies dataset: This multi-criteria dataset includes user ratings for movies based on various aspects such as direction, story, acting, and visuals. It provides a complementary domain for evaluating the proposed model, enabling an analysis of its generalizability beyond TEL. By leveraging this dataset, we demonstrate how the DeepFM-SVD + + model effectively captures complex relationships in user preferences across different domains.

The inclusion of these datasets ensures a comprehensive evaluation of the model's capabilities in both educational and entertainment contexts. The ITM-REC dataset highlights the model's relevance to technology-enhanced learning, while the Yahoo Movies dataset showcases its adaptability to other domains, as shown in Table 1.

TEL multicriteria dataset analysis

The experiment was conducted using an educational dataset collected from questionnaires for five (5) years, from 2017 to 2022. The ITM-Rec[34] dataset was collected from graduate students enrolled in data management and analytics specialization at the ITM department at the Illinois Institute of Technology[34]. Students' individual preference on final project topics in three courses, Data Analytics (DA), Data Science (DS), and Database (DB), was collected. The students provided ratings on their selection by giving an overall rating (c_0) and additional ratings for three criteria: App (c_1), Data (c_2), and Ease (c_3). The App represents the likeness of the application domain, the Data represents the likeness of the data's processing or storage, and Ease represents the student's likeness of the degree of ease in using the data for the final project. The dataset contains 5,230 ratings given by 476 unique users over 70 unique items. Ratings of each criterion were measured on a scale of 5, varying from 1 to 5, with 1 being the lowest preference and 5 being the highest preference. A sample of the dataset is shown in Table 2. While analyzing the data, it was observed that a student could provide ratings for an item more than once based on the class in which the topic was taken. For example, student A can take item B in both DA and DM classes and provide different ratings based on the classes in which the item was taken. This led to duplicates in the dataset, as there was no unique user-item interaction. To resolve this, we created a new column named UID. The values in the UID column were generated based on the combination of the UserID, ItemID, and Class, using a dash as a separator. The UID was used to identify each user-item interaction uniquely.

Furthermore, the dataset was cleaned to check and remove inconsistencies to improve its quality. The average rating for the App, Data, Ease, and Overall ratings are 3.421, 3.390, 3.177, and 3.374, respectively. Although the statistical analysis in Table 3 shows that all the aspect ratings had positive correlations with the overall ratings, it

	Overall	App	Data	Ease
Overall	1.000	**0.799**	**0.725**	**0.582**
App	**0.799**	1.000	0.776	0.629
Data	**0.725**	0.776	1.000	0.662
Ease	**0.582**	0.629	0.662	1.000

Table 3. Pearson correlation matrix of the ITM-Rec dataset. Significant values are in bold.

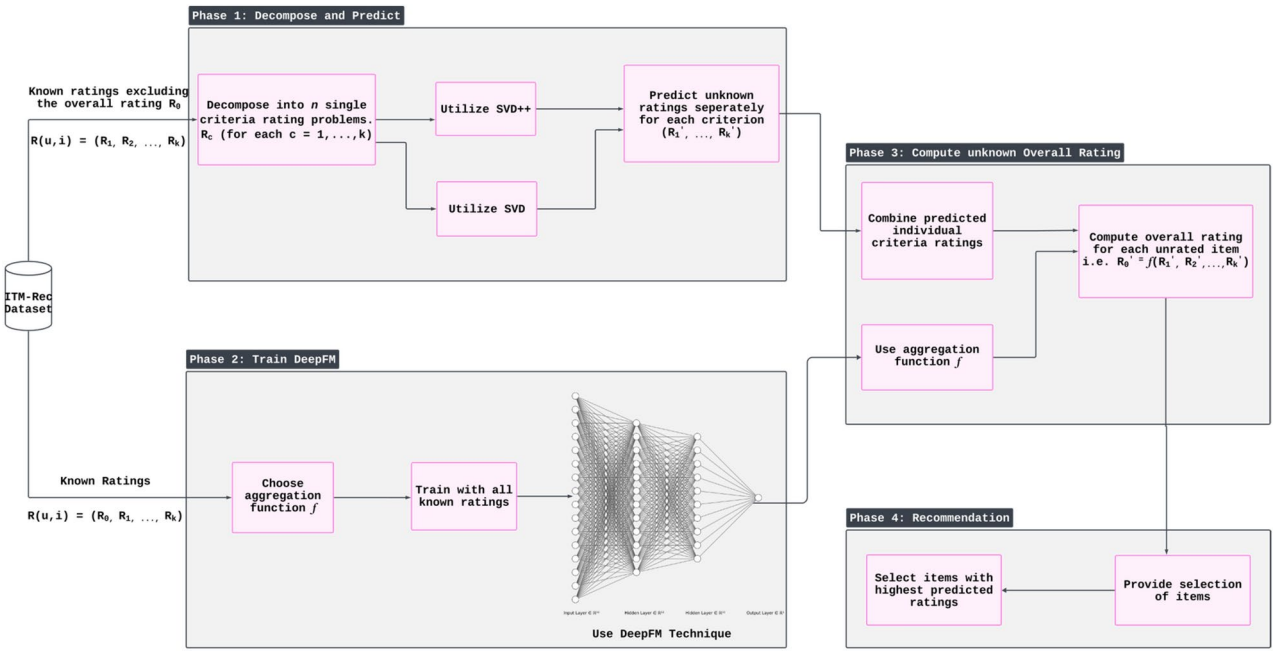

Fig. 5. Proposed DeepFM-based multi-criteria recommender systems framework.

can be seen from the table that the App aspect had the most significant influence in deciding the overall ratings, while the Ease had the least impact on the overall ratings.

Proposed deep learning-based MCRSs framework

DeepFM combines FM for low-order feature interactions with a DNN for high-order dependencies, enabling it to capture both explicit and complex non-linear relationships. In this hybrid structure, the FM component explicitly models pairwise feature interactions, while the DNN component learns intricate, higher-order patterns that traditional methods like SVD++ cannot fully address. By unifying these two approaches, DeepFM-SVD++ gains a better understanding of multi-dimensional learner preferences in TEL, leading to improved predictive performance and greater recommendation accuracy in multi-criteria contexts. Additionally, the ability to learn from limited data (via embeddings) helps mitigate cold-start and data sparsity issues more effectively than purely factorization-based models. Our proposed aggregation function approach is structured into four key phases, as outlined below:

- Phase 1: Decompose the multi-criteria rating dataset from n-dimensional rating problems into n-single criteria rating problems. To predict the known ratings with UserID and ItemID, we used two traditional collaborative filtering techniques, which are SVD and SVD++.
- Phase 2: In this phase, the aggregation function f is chosen using the DeepFM technique. The known overall rating, considered as another criterion, is trained using a DeepFM algorithm.
- Phase 3: The overall rating of each unrated item is computed on the n predicted individual criteria ratings and the aggregation function f.
- Phase 4: This is the recommendation phase. The calculated predictions are used to support the user's decision. The user is recommended a set of items with the highest predicted ratings. Figure 5 summarizes the working process of our proposed approach.

Implementation of the proposed approach

The proposed system was implemented using Python. Python produces an exceptional amount of power and versatility in deep learning environments. In the implementation of our DeepFM model, we utilized various Python libraries, including TensorFlow, Keras, and Scikit-learn. These libraries collectively enabled us to build,

train, and evaluate the model, harnessing the capabilities of both DNN and traditional machine learning techniques. For our proposed model, we configured the embedding dimension to be 10 and introduced L2 regularization with a strength of 0.0001. Our Dense layer consisted of 128 units, offering a balance between model complexity and computational efficiency.

To optimize the model, we employed the Adam optimizer with a learning rate set to 0.01. In the training phase, the model was updated using mini-batches, each comprising 32 samples. Our model underwent training for a maximum of 45 epochs, during which we used two callbacks: 'ReduceLROnPlateau' and 'EarlyStopping' to prevent overfitting and improve convergence. A sigmoid activation function was used for the output layer, which scales the model's predictions to the range [0,1]. Subsequently, we transformed these values to fit the ITM-Rec rating scale of [1,5] using a lambda layer. For robust evaluation, we conducted cross-validation using K-fold, which determines how many subsets the data is partitioned into for training and validation. Furthermore, we incorporated bias terms, which are crucial for capturing user and item biases. These biases were separately represented as embedding layers within the model.

In the implementation of SVD++ and SVD, we harnessed the power of the Surprise library. We initiated a reader object, explicitly specifying the rating scale within the range of 1 to 5. We then defined a parameter grid tailored for hyperparameter tuning, encompassing three hyperparameters: the number of training epochs (n_epochs: 20, 30, 40), the learning rate (lr_all: 0.005, 0.01, 0.05), and the regularization parameter (reg_all: 0.06, 0.1, 0.2, 0.5). To perform hyperparameter tuning, we employed GridSearchCV, configuring it with the algorithms and the predefined parameter grid. Subsequently, we fitted the grid search to our training data, allowing it to explore various combinations of hyperparameters. As a result, we obtained the best parameters and the corresponding best estimator (model). Taking the best estimator, we proceeded to train it on our training set, enabling the model to capture patterns and relationships within the data. Subsequently, we employed this trained model to make predictions on our test set, offering valuable insights into its predictive capabilities.

Performance evaluation metrics

We evaluate the recommendation's accuracy and performance based on the top-N recommendation task. We broadly used six different evaluation metrics to measure the performance accuracy, which are as follows:

1. Mean absolute error (MAE): The MAE was used to measure the rating prediction accuracy of our proposed recommendation approach. MAE estimates the deviation between the predicted ratings and the actual ratings. Equation 11 shows how MAE is calculated, where n is the number of test set ratings generated from the test sets, x_j' and x_j are the predicted and actual ratings, respectively.

$$MAE = \frac{1}{n}\sum_{j=1}^{n}|x_j\prime - x_j| \tag{11}$$

2. Root mean square error (RMSE): RMSE was also used to measure the rating prediction accuracy of our proposed system. RMSE uses the squared standard deviation and emphasizes larger errors. Equation 12 shows how RMSE is calculated, where n is the number of test set ratings generated from the test sets, x_j' and x_j are the predicted and actual ratings, respectively.

$$RMSE = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(x_j\prime - x_j)^2} \tag{12}$$

3. F1-score: This was used to measure the usage prediction, and it is the harmonic mean of the precision and recall. Precision is the fraction of true positive predictions over the total number of positive predicted recommendations. Recall is the fraction of the positive predictions to a specific user's list of relevant items[25]. Equations 13, 14, and 15 show how the precision, recall, and F1-score are calculated, respectively.

$$Precision = \frac{Numberoftruepositive}{Numberoftruepositive + Numberoffalsepositive} \tag{13}$$

$$Recall = \frac{Numberoftruepositive}{Numberoftruepositive + Numberoffalsenegative} \tag{14}$$

$$F1 - score = 2 \times \frac{precision \times recall}{precision + recall} \tag{15}$$

4. Mean average precision (MAP): MAP was also used to measure the usage prediction; it computes the Average Precision (AP) across different levels of recall. Equations 16 and 17 show how the AP and MAP are calculated. Where N is the total number of relevant items among the list of recommended items.

$$AP = \frac{\sum_{j=1}^{n} precision\,(j) \times recall(j)}{numberofrelevantitems} \tag{16}$$

$$MAP = \frac{\sum_{i=1}^{N} AP_i}{N} \tag{17}$$

5. Area under the curve (AUC): This was used to measure the ranking accuracy. AUC of a receiver operating characteristics (ROC) curve measures how accurately the proposed algorithm separates predictions into relevant and irrelevant by finding the sensitivity curve against specificity. Equation 18 shows how the AUC_u is computed, where $rank_{ui}$ is the position of the *nth* relevant item among a list of *N* recommended items, and t_{pu} is the number of true positives of each user.

$$AUC_u = \frac{1}{N}\left[\left(\sum_{i=1}^{tp_u} rank_{ui}\right) + \left(\frac{tp_u + 1}{2}\right)\right] \tag{18}$$

6. Fraction of concordant pairs (FCP): This was also used to measure the ranking accuracy of our proposed system. The n_c is the number of concordant pairs calculated as $\sum_{u \in U} |i, j| \{r_{ui}\prime > r_{uj}\prime \Rightarrow r_{ui} > r_{uj}\}$, and n_d is the corresponding number of discordant pairs calculated as $\sum_{u \in U} |i, j| \{r_{ui}\prime > r_{uj}\prime \Rightarrow r_{ui} \leq r_{uj}\}$. Concordant pairs are predicted ratings $r_{ui}\prime$ and $r_{uj}\prime$ for some items *i* and *j*, such that if $r_{ui}\prime > r_{uj}\prime$, then the corresponding ratings r_{ui} and r_{uj} from the dataset must also satisfy the same condition, i.e., $r_{ui} > r_{uj}$, else the items *i* and *j* are discordant pairs. FCP is calculated as shown in Eq. 19.

$$FCP = \frac{a_c}{a_c - a_d} \tag{19}$$

Results and discussion

We conducted multiple experiments to evaluate our recommendation models. First, we partitioned the dataset into training and test sets using K-fold cross-validation, exploring two values of K (5 and 10) and varying Top-N recommendations (10 and 20). This resulted in four distinct configurations: *Top-10 with fivefold, Top-20 with fivefold, Top-10 with tenfold, and Top-20 with tenfold*. To assess the effectiveness of our aggregation function-based DeepFM_SVD++and DeepFM_SVD models, we used a range of evaluation metrics detailed in Section "Performance evaluation metrics".

We then compared DeepFM-SVD++with baseline models, including SVD and SVD++. As shown in Table 4, our model consistently outperformed traditional matrix factorization techniques, highlighting its ability to capture complex multi-criteria relationships, particularly in sparse datasets like ITM-Rec.

Our findings demonstrate that DeepFM_SVD++_MCRS and DeepFM_SVD_MCRS significantly outperform SVD++and SVD. The excellent performance of DeepFM-SVD++over SVD and SVD++stems from its hybrid architecture, which combines deep neural networks for capturing non-linear interactions and factorization components for low-order feature interactions. By modeling both explicit and implicit feedback, the approach mitigates data sparsity and cold-start issues more effectively than traditional methods. Our experiments confirm that DeepFM-SVD++consistently achieves lower RMSE and MAE. These reductions reflect a better model fit, resulting in greater accuracy when predicting user preferences. The substantial

Settings	Algorithms	RMSE	MAE	Precision	Recall	F1-score	MAP	AUC	FCP
5-Fold & Top-10	*SVD++*	1.2837	1.0501	0.6938	0.8497	0.7639	0.7814	0.7459	0.6899
	DeepFM_SVD++_MCRS	**0.8923**	**0.7232**	**0.7874**	**0.9468**	**0.8598**	**0.9238**	**0.9145**	**0.8410**
	SVD	1.2875	1.0277	0.6874	0.8409	0.7565	0.7967	0.7545	0.6907
	DeepFM_SVD_MCRS	0.9114	0.7367	0.7781	0.9383	0.8507	0.9150	0.9075	0.8367
5-Fold & Top-20	*SVD++*	1.2890	1.0548	0.6922	0.8490	0.7626	0.7803	0.7443	0.6881
	DeepFM_SVD++_MCRS	**0.8921**	**0.7230**	**0.7874**	**0.9452**	**0.8591**	**0.9236**	**0.9142**	**0.8410**
	SVD	1.2906	1.0311	0.6872	0.8406	0.7562	0.7965	0.7536	0.6895
	DeepFM_SVD_MCRS	0.9112	0.7366	0.7781	0.9366	0.8500	0.9147	0.9072	0.8366
10-Fold & Top-10	*SVD++*	1.2859	1.0500	0.6935	0.8446	0.7616	0.7836	0.7448	0.6903
	DeepFM_SVD++_MCRS	**0.8979**	**0.7257**	**0.7605**	**0.9640**	**0.8502**	**0.9251**	**0.9176**	**0.8534**
	SVD	1.2908	1.0420	0.6779	0.8545	0.7560	0.7937	0.7447	0.6875
	DeepFM_SVD_MCRS	0.9180	0.7351	0.7582	0.9468	0.8421	0.9195	0.9084	0.8383
10-Fold & Top-20	*SVD++*	1.2917	1.0549	0.6923	0.8456	0.7613	0.7822	0.7428	0.6881
	DeepFM_SVD++_MCRS	**0.8975**	**0.7254**	**0.7608**	**0.9640**	**0.8505**	**0.9248**	**0.9172**	**0.8533**
	SVD	1.2936	1.0445	0.6778	0.854	0.7558	0.7558	0.7437	0.6861
	DeepFM_SVD_MCRS	0.9176	0.7348	0.7586	0.9469	0.8423	0.9193	0.9082	0.8383
Average	*SVD++*	1.2876	1.0525	0.6930	0.8472	0.7624	0.7819	0.7445	0.6891
	DeepFM_SVD++_MCRS	**0.8950**	**0.7243**	**0.7740**	**0.9550**	**0.8549**	**0.9243**	**0.9159**	**0.8472**
	SVD	1.2906	1.0363	0.6826	0.8475	0.7561	0.7857	0.7491	0.6885
	DeepFM_SVD_MCRS	0.9146	0.7358	0.7683	0.9422	0.8463	0.9171	0.9078	0.8375

Table 4. Evaluation results from ITM-Rec dataset. Significant values are in bold.

	RMSE	*MAE*	*Precision*	*Recall*	*F1-score*	*MAP*	*AUC*	*FCP*
Difference (SVD++)	0.3926	0.3281	0.0811	0.1078	0.0926	0.1425	0.1714	0.1581
% Improvement (SVD++)	30%	31%	12%	13%	12%	18%	23%	23%
Difference (SVD)	**0.3957**	**0.3120**	**0.0915**	**0.1075**	**0.0988**	**0.1387**	**0.1668**	**0.1587**
% Improvement (SVD)	**31%**	**30%**	**13%**	**13%**	**13%**	**18%**	**22%**	**23%**

Table 5. Level of accuracy improvement between DeepFM SVD++_MCRS and the traditional RS technique. Significant values are in bold.

	RMSE	*MAE*	*Precision*	*Recall*	*F1-score*	*MAP*	*AUC*	*FCP*
Difference (SVD++)	0.3730	0.3167	0.0753	0.0949	0.0839	0.1353	0.1634	0.1484
% Improvement (SVD++)	29%	30%	11%	11%	11%	17%	22%	22%
Difference (SVD)	0.3761	0.3005	0.0857	0.0946	0.0902	0.1315	0.1587	0.1490
% Improvement (SVD)	29%	29%	13%	11%	12%	17%	21%	22%

Table 6. Level of accuracy improvement between DeepFM SVD_MCRS and the traditional RS technique.

Model	*RMSE*	*MAE*	*Precision*	*Recall*	*F1-score*	*MAP*	*AUC*	*FCP*
DeepFM SVD++_MCRS	**30.49%**	**31.18%**	**11.70%**	**12.72%**	**12.14%**	**18.22%**	**23.03%**	**22.94%**
DeepFM SVD_MCRS	**29.14%**	**29.00%**	**12.55%**	**11.17%**	**11.92%**	**16.73%**	**21.18%**	**21.65%**
Yong and David [21]	-	-	-	-	6.4%	-	-	-
Yong et al. [25]	-	-	-	-	9.6%	-	-	-

Table 7. Comparison of percentage improvement with related works. Significant values are in bold.

performance gains observed in precision, recall, and F1-score stem from DeepFM-SVD++'s ability to capture non-linear feature dependencies dynamically. This interaction mechanism enables the model to provide more personalized recommendations, addressing the issue of over-specialization commonly found in traditional matrix factorization-based approaches. Additionally, the higher MAP, FCP, and AUC values confirmed enhanced ranking performance and more substantial discriminative capability, reinforcing the benefit of multi-criteria and deep learning-based optimizations in TEL scenarios.

To assess overall accuracy improvements across different folds and Top-N recommendation settings, we averaged each technique's performance. Tables 5 and 6 present the magnitude of improvement for DeepFM_SVD++and DeepFM_SVD MCRS on every metric, computed as the absolute difference relative to the baseline techniques. We then derived the percentage improvement by dividing this difference by the baseline performance. As an example, Eq. 20 illustrates the percentage improvement between SVD++and DeepFM_SVD++.

$$percentimprovement = \frac{|DeepFM_SVD++_MCRS - SVD++|}{SVD++} \times 100 \tag{20}$$

In summary, our proposed DeepFM-based models consistently outperformed traditional techniques across all evaluation metrics, with DeepFM_SVD++_MCRS achieving the best overall performance. This model delivered more accurate predictions, an enhanced ranking of recommended items, and a better balance between precision and recall compared to SVD and SVD++. Additionally, DeepFM_SVD_MCRS also demonstrated excellent performance over traditional methods, with results closely approaching those of DeepFM_SVD++_MCRS, further validating the effectiveness of our proposed multi-criteria approach.

The efficiency of our proposed model is further evaluated by comparing it with previous works, as presented in Table 7. Direct comparison with prior studies is challenging due to variations in datasets and model architectures, as most TEL studies employ single-rating techniques. To ensure a meaningful evaluation, we selected two studies that utilized the ITM-Rec multi-criteria dataset and assessed performance based on percentage improvement over baseline algorithms, following the same approach used in Tables 5 and 6.

Figure 6 displays a correlation graph depicting the relationship between actual ratings and predicted ratings for each of the algorithms. DeepFM_SVD++_MCRS, as shown in Fig. 6a, achieved a notably high correlation coefficient of 0.777, signifying a strong positive correlation between the actual and predicted ratings. This outcome highlights the model's ability to forecast user ratings for items, demonstrating its effectiveness accurately. Similarly, DeepFM_SVD_MCRS, illustrated in Fig. 6d, exhibited a better correlation coefficient of 0.769, which further emphasizes its predictive accuracy. In contrast, the traditional SVD++and SVD algorithms yielded relatively lower correlation coefficients of 0.458 and 0.465, as depicted in Figs. 6b, c, respectively, and these lower coefficients suggest a less accurate prediction.

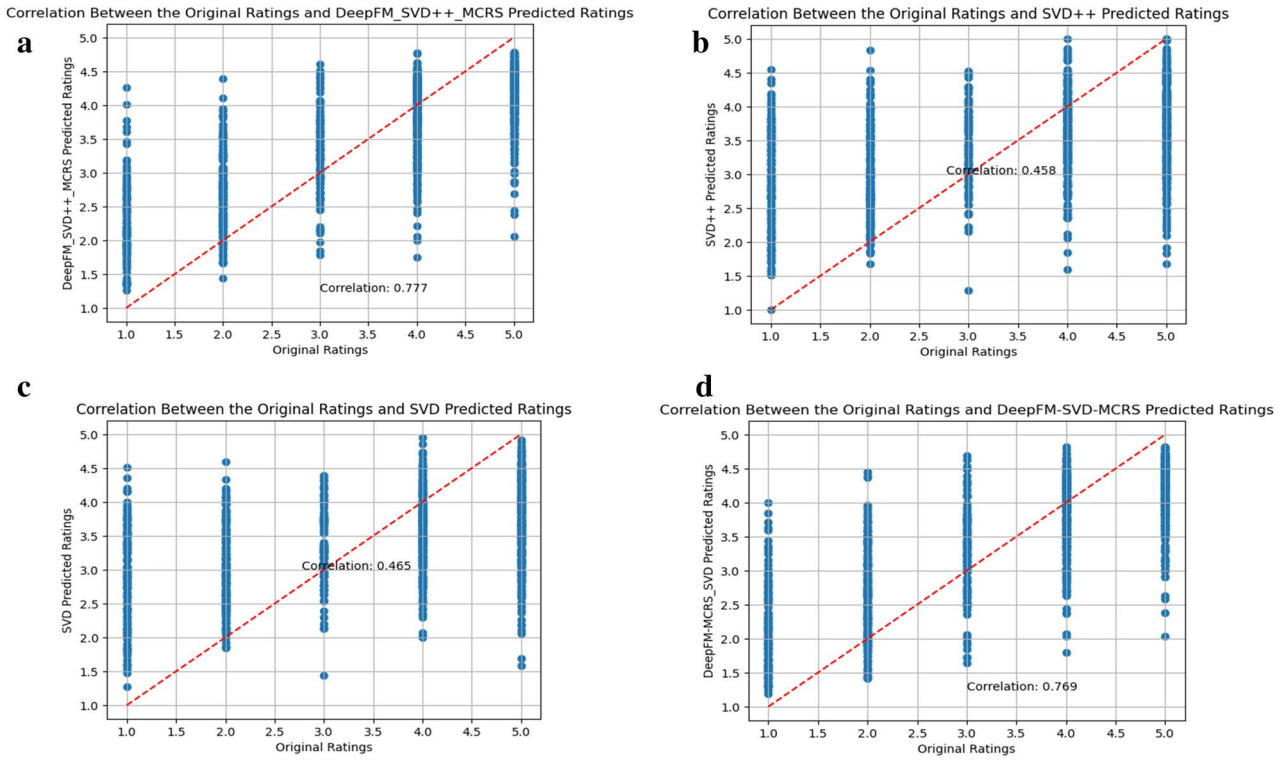

Fig. 6. Correlation graphs between Actual ratings and Predicted ratings for each of the algorithms. (**a**) Actual ratings vs. DeepFM_SVD + + _MCRS Predicted ratings. (**b**) Actual ratings vs. SVD + + Predicted ratings. (**c**) Actual ratings vs. SVD Predicted ratings (**d**) Actual ratings vs. DeepFM_SVD_MCRS Predicted ratings.

Extension

In this study, we propose an aggregation function-based DeepFM model for MCRSs within TEL. To demonstrate the model's generalization across domains, we conducted additional experiments using the Yahoo Movies dataset, a multi-criteria rating dataset with 62,156 ratings from 6,078 users across 976 movies and a sparsity of 99.0%. Users rated movies based on five criteria: direction (cr1), action (cr2), story (cr3), visuals (cr4), and an overall rating (cr0). The original letter-grade ratings (A + to F) were transformed to a numeric scale of 1 to 13, where A + indicates the highest preference.

For evaluation, we applied K-fold cross-validation (5 and 10 folds) and maintained consistent configurations for SVD and SVD + + from the ITM-Rec experiments. Experiments were conducted for Top-10 and Top-20 recommendations, highlighting the model's scalability and robust performance across domains.

Tables 8, 9, and 10 present the evaluation outcomes derived from our experiments. The results manifestly showcase the generalizability of our proposed aggregation function-based DeepFM model across all evaluation metrics, significantly surpassing the performance of the baseline algorithms, with accuracy improvements spanning between 9 and 40%. Table 11 shows the comparison of our proposed model were compared with studies that utilized the Yahoo Movies dataset, using percentage impact analysis consistent with the comparisons in Tables 9 and 10. Further analysis was conducted to examine the correlation between the actual ratings and the predicted ratings generated by each algorithm, as shown in Figs. 7a, b, c, and d. Our proposed DeepFM approach exhibited a strong correlation with the actual ratings, reflected by correlation coefficients of 0.912 and 0.908. In contrast, the baseline algorithms (SVD and SVD + +) only achieved a correlation coefficient of 0.623 each, indicating less correlation with the actual ratings.

Conclusion and future work

In this study, we introduce an MCRS that leverages deep learning techniques. Our model employs the DeepFM architecture to learn an aggregation function. Once trained, the DeepFM model collaborates with conventional single-criterion rating mechanisms to predict unknown overall ratings. Based on these predictions, we then generate a Top-N recommendation list. For our MCRS, we integrate traditional matrix factorization techniques, specifically SVD and its extension, SVD + +. Each of these methods is paired with DeepFM to enhance the recommendation capabilities. We carried out experiments to validate the performance of our proposed DeepFM-based aggregation function. For benchmarking, we compared our results with those of leading contemporary algorithms. Across multiple datasets, our proposed model consistently surpassed the baseline techniques and similar related works. The notable improvement in the prediction accuracy, ranking accuracy, classification accuracy, and correlation with actual ratings confirmed the effectiveness of our approach. We also

Settings	Algorithms	RMSE	MAE	Precision	Recall	F1-score	MAP	AUC	FCP
5-Fold & Top-10	*SVD++*	2.7035	1.9945	0.8008	0.8810	0.8390	0.8904	0.8303	0.7459
	DeepFM_SVD++_MCRS	**1.6492**	**1.2015**	**0.8779**	**0.9664**	**0.9201**	**0.9751**	**0.9595**	**0.8955**
	SVD	2.7107	2.0248	0.8036	0.8748	0.8377	0.8929	0.8292	0.7478
	DeepFM_SVD_MCRS	1.6583	1.2104	0.8765	**0.9676**	0.9198	0.9745	0.9592	**0.8963**
5-Fold & Top-20	*SVD++*	2.7047	1.9988	0.7990	0.8755	0.8355	0.8883	0.8313	0.7478
	DeepFM_SVD++_MCRS	**1.6508**	**1.2038**	**0.8769**	**0.9651**	**0.9189**	**0.9745**	**0.9599**	**0.8967**
	SVD	2.7122	2.0297	0.8018	0.8703	0.8346	0.8909	0.8308	0.7498
	DeepFM_SVD_MCRS	1.6612	1.2136	0.8749	**0.9662**	0.9183	0.9739	0.9597	**0.8977**
10-Fold & Top-10	*SVD++*	2.6952	1.9962	0.8022	0.8825	0.8405	0.8896	0.8297	0.7471
	DeepFM_SVD++_MCRS	**1.6521**	**1.2136**	**0.8822**	**0.9640**	**0.9213**	**0.9747**	**0.9595**	**0.8961**
	SVD	2.7145	2.0280	0.8031	0.8731	0.8366	0.8926	0.8299	0.7484
	DeepFM_SVD_MCRS	1.6571	1.2222	0.8795	**0.9654**	0.9204	**0.9749**	0.9596	**0.8966**
10-Fold & Top-20	*SVD++*	2.6969	1.9996	0.8007	0.8767	0.8370	0.8877	0.8311	0.7492
	DeepFM_SVD++_MCRS	**1.6530**	**1.2156**	**0.8810**	**0.9628**	**0.9201**	**0.9741**	**0.9599**	**0.8972**
	SVD	2.7202	2.0351	0.8005	0.8677	0.8328	0.8903	0.8305	0.7500
	DeepFM_SVD_MCRS	1.6591	1.2249	0.8782	**0.9644**	0.9192	0.9743	0.9599	**0.8980**
Average	*SVD++*	2.7001	1.9973	0.8007	0.8789	0.8380	0.8890	0.8306	0.7475
	DeepFM_SVD++_MCRS	**1.6513**	**1.2086**	**0.8795**	**0.9646**	**0.9201**	**0.9746**	**0.9597**	**0.8964**
	SVD	2.7144	2.0294	0.8023	0.8715	0.8354	0.8917	0.8301	0.7490
	DeepFM_SVD_MCRS	1.6589	1.2178	0.8773	**0.9659**	0.9194	0.9744	0.9596	**0.8972**

Table 8. Evaluation results from yahoo movies. Significant values are in bold.

	RMSE	*MAE*	*Precision*	*Recall*	*F1-score*	*MAP*	*AUC*	*FCP*
Difference (SVD++)	1.0488	0.7887	0.0788	0.0857	0.0821	0.0856	0.1291	0.1489
% Improvement (SVD++)	**39%**	**39%**	**10%**	**10%**	**10%**	**10%**	**16%**	**20%**
Difference (SVD)	1.0631	0.8208	0.0772	0.0931	0.0847	0.0829	0.1296	0.1474
% Improvement (SVD)	**39%**	**40%**	**10%**	**11%**	**10%**	**9%**	**16%**	**20%**

Table 9. Level of accuracy improvement between DeepFM SVD++_MCRS and the traditional RS technique. Significant values are in bold.

	RMSE	*MAE*	*Precision*	*Recall*	*F1-score*	*MAP*	*AUC*	*FCP*
Difference (SVD++)	1.0412	0.7795	0.0766	0.0870	0.0814	0.0854	0.1290	0.1497
% Improvement (SVD++)	**39%**	**39%**	**10%**	**10%**	**10%**	**10%**	**16%**	**20%**
Difference (SVD)	1.0555	0.8116	0.0750	0.0944	0.0840	0.0827	0.1295	0.1482
% Improvement (SVD)	**39%**	**40%**	**9%**	**11%**	**10%**	**9%**	**16%**	**20%**

Table 10. Level of accuracy improvement between DeepFM SVD_MCRS and the traditional RS technique. Significant values are in bold.

Model	*RMSE*	*MAE*	*Precision*	*Recall*	*F1-score*	*MAP*	*AUC*	*FCP*
DeepFM SVD++_MCRS	**38.84%**	**39.49%**	**9.84%**	**9.74%**	**9.80%**	**9.63%**	**15.54%**	**19.92%**
DeepFM SVD_MCRS	**38.88%**	**39.99%**	**9.35%**	**10.84%**	**10.05%**	**9.28%**	**15.60%**	**19.78%**
Qusai[5]	26.27%	26.20%	-	-	-	-	-	-
Son-Lam and Quang-Hung[20]	5.15%	7.12%	-	-	-	-	-	-
Nour et al.[23]	-	6.51%	-	-	8.94%	14.65%	-	5.08%

Table 11. Comparison of percentage improvement with related works. Significant values are in bold.

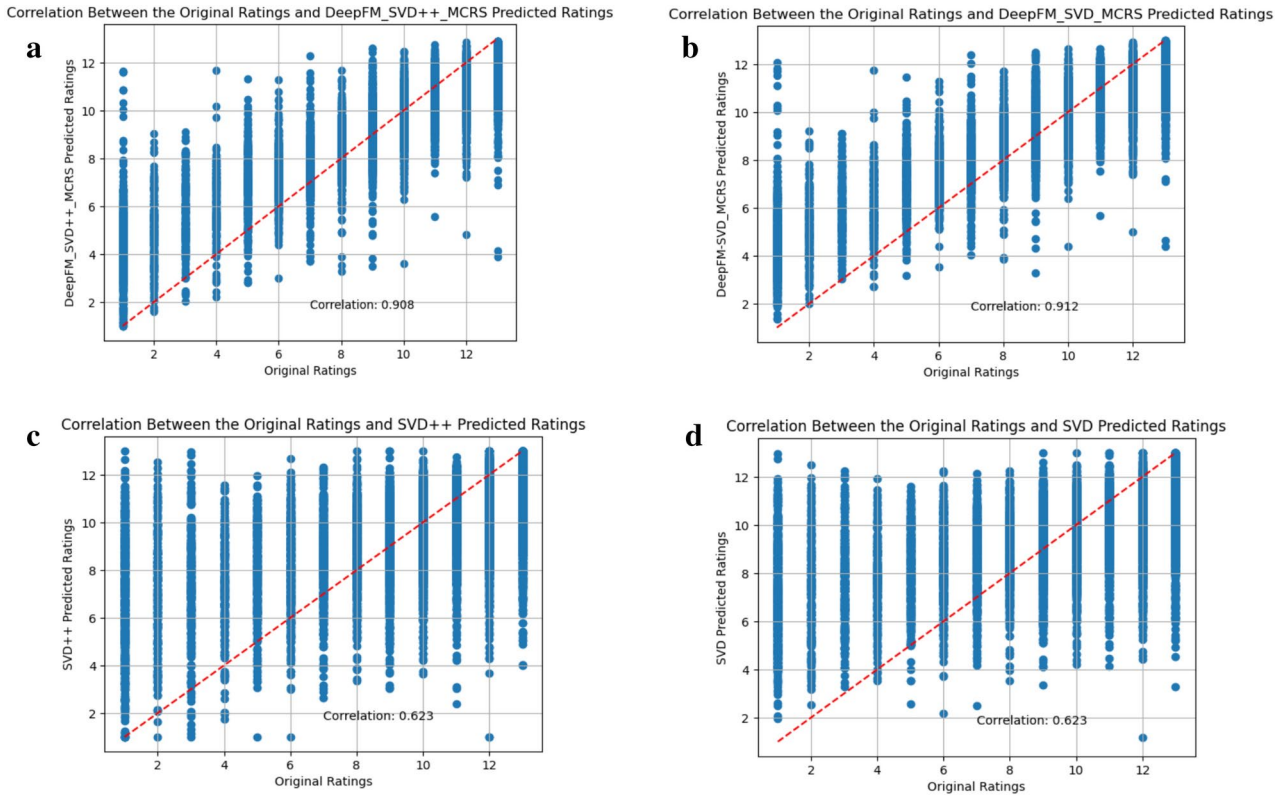

Fig. 7. Correlation graphs between actual ratings and predicted ratings for each of the algorithms. (**a**) Actual ratings vs. DeepFM_SVD + + _MCRS Prpdicted ratings. (**b**) Actual ratings vs. DeepFM_SVD_MCRS predicted ratings (**c**) Actual ratings vs. SVD + + Prpdicted ratings (**d**) Actual ratings vs. SVD predicted ratings.

extended our proposed approach to the entertainment domain, using the Yahoo Movies dataset. Again, our model demonstrated evident improvements, showing its adaptability and generalizability across domains.

While this study has demonstrated the effectiveness of the proposed DeepFM-SVD + + model using the ITM-REC and Yahoo Movies datasets, there remains considerable potential for advancing research in MCRSs for TEL, a field still in its early stages. Future research could explore the following areas:

(1) Evaluation on large-scale MOOC datasets: Future experiments could focus on applying the proposed model to large-scale MOOC datasets such as Coursera, edX, and Udemy. These datasets can provide a more diverse and real-world evaluation of the model's scalability and robustness across different learning environments.
(2) Context-aware mechanics and advanced deep learning models: Integrating context-aware mechanics into MCRSs represents a promising avenue for improving the quality of recommendations. For example, contextual factors like learner preferences, location, and time could be leveraged to deliver personalized recommendations in the TEL domain. Additionally, extending and adapting advanced deep learning models such as Deep & Cross Network (DCN), Deep Interest Network (DIN), and xDeepFM for multi-criteria recommendation tasks in TEL could provide new insights into feature interactions and user modeling.
(3) Ablation study: To further understand the contributions of individual components, a future ablation study will help evaluate the performance of the proposed model by excluding or altering components such as DeepFM and SVD + + . This analysis will provide better insights into the relative importance of each element in optimizing multi-criteria relationships.

These future directions aim to extend the DeepFM-SVD + + model's applicability and robustness and contribute to the broader development of MCRSs, advancing their role in addressing the unique challenges of TEL environments.(Supplementary information)

Data availability

The Educational dataset used and analyzed during this study were cited and are open-sourced (https://www.kaggle.com/datasets/irecsys/itmrec). The yahoo movies data is provided within the manuscript or supplementary information files.

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Author contributions

L.S. did the conceptualization, writing the original draft, preparation, methodology, data curation, discussion and conclusion, and future work of the manuscript. H. M. reviewed and Edited the manuscript; Y. A. performed formal analysis, reviewing and editing of the manuscript; and H. M. reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to L.S.

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